

Matsilele A. Mabaso

Senior Data Scientist | Data Engineer |
ML/AI Specialist

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Professional Summary

Senior Data Scientist and ML Engineer with 8+ years of experience designing, building and deploying production-grade machine learning, analytics and data engineering solutions across financial services, manufacturing and research environments. PhD in Electrical Engineering specialising in Computer Vision and Machine Learning, with strong foundations in mathematics, statistics, software engineering and data engineering. Experienced across the end-to-end ML lifecycle, from data preparation, feature engineering and model development through API-based inference, Docker containerisation, CI/CD and Azure cloud deployment. Hands-on experience delivering production ML applications using Django/Django REST Framework, React, GitHub Actions, Azure Container Registry (ACR), Azure Kubernetes Service (AKS), Azure Blob Storage, Databricks, Apache Spark and MLflow. Focused on building reliable, reproducible and scalable ML systems that translate models into operational business applications.

Core Competences

ML Engineering & MLOps	Production ML systems, ML lifecycle management, ML pipelines, model deployment, monitoring, retraining, MLflow, CI/CD, Docker, Kubernetes, model governance
Azure ML Platform	Azure, Azure Kubernetes Service (AKS), Azure Container Registry (ACR), Azure Blob Storage, Databricks, Apache Spark (PySpark), Delta Lake
Machine Learning	Regression, classification, clustering, deep learning, NLP, predictive analytics, feature engineering, model evaluation, statistical modelling
Programming	Python, SQL, R, PyMC, JavaScript, TypeScript, React, Django, Django REST Framework
Inference & Applications	REST APIs, production ML APIs, batch inference, model serving, Django backends, React frontends, Streamlit
CI/CD & DevOps	Git, GitHub, GitHub Actions, GitLab, Docker, Azure Container Registry, Kubernetes/AKS, deployment automation
Generative AI	LLMs, RAG systems, embeddings, FAISS, ChromaDB, semantic search, private LLM applications
Databases & BI	PostgreSQL, MySQL, MongoDB, vector search, Power BI, Streamlit, Matplotlib, Seaborn, ggplot
Leadership	Technical leadership, mentoring, stakeholder engagement, cross-functional collaboration, Scrum, sprint planning

Education

2022	Data Engineering , <i>Explore Data Science Academy</i>
2015–2019	Doctor of Electrical Engineering (Computer Vision & Machine Learning) , <i>University of Johannesburg</i>
2012–2014	Master of Electrical Engineering (Image Processing) , <i>University of Johannesburg</i> , Distinction
2010	Honours in Mathematical Statistics , <i>University of Limpopo</i> , Cum Laude
2007–2009	BSc Mathematics & Statistics , <i>University of Limpopo</i> , Cum Laude

Professional Experience

Nov 2024 – **Senior Data Scientist**, HULAMIN, Pietermaritzburg, South Africa

- Present
- Design, build and deploy production-grade machine learning applications across the manufacturing environment, taking ML solutions from model development through API development, containerisation, CI/CD and cloud deployment.
 - Develop scalable ML pipelines using Python, SQL, PySpark, Apache Spark and Databricks for operational data processing, feature engineering and machine learning workloads.
 - Build production backends using Django and Django REST Framework to expose machine learning models through REST APIs, and develop React/TypeScript frontends that deliver predictions and analytics to business users.
 - Containerise ML applications using Docker and implement GitHub-based CI/CD workflows with GitHub Actions to automate build, test and deployment processes.
 - Deploy containerised ML applications on Microsoft Azure using Azure Container Registry (ACR) for image management and Azure Kubernetes Service (AKS) for production workloads.
 - Implement cloud-based model artifact management using Azure Blob Storage, including storage and retrieval of trained Python pickle (.pkl) model artifacts for production inference.
 - Operationalise the ML lifecycle through reproducible pipelines, model evaluation, deployment, monitoring and retraining, integrating data, models, APIs and application interfaces into complete production systems.
 - Lead cross-functional analytics and ML initiatives, translating complex technical requirements into reliable solutions for engineering and operations stakeholders.
 - Mentor data analysts and engineers in statistical modelling, machine learning, experimentation, software practices and the end-to-end ML lifecycle.

Jan 2022 – Oct 2024 **Data Scientist**, SASOL, Sandton, South Africa

- Led end-to-end data science projects improving operational efficiency through predictive modelling and statistical analysis.
- Built scalable ML pipelines supporting predictive maintenance and process optimisation.
- Designed data-driven evaluation frameworks to assess the impact of operational interventions and process improvements.
- Developed automated monitoring dashboards enabling real-time tracking of key operational metrics.
- Designed and deployed a time-series forecasting API using Django and React.js, integrating statistical and ML-based forecasting models.
- Presented insights and technical recommendations to engineering, operational and executive stakeholders.

Nov 2019 – Dec 2021 **Machine Learning Engineer**, CONSUMER PROFILE BUREAU, Bryanston, South Africa

- Developed ML models for credit scoring and risk assessment, improving predictive accuracy by 12%.
- Automated analytics workflows with R Shiny dashboards, reducing manual reporting by 60%.
- Engineered predictive models supporting real-time contact-centre decision making.
- Partnered with stakeholders to interpret model outcomes and translate ML outputs into business applications.

May 2018 – Oct 2019 **Researcher (Computer Vision & ML)**, COUNCIL FOR SCIENTIFIC AND INDUSTRIAL RESEARCH (CSIR), Pretoria, South Africa

- Conducted applied ML research in image processing and robotic perception.
- Designed and implemented machine learning and computer-vision algorithms for real-world research problems.
- Published in peer-reviewed journals and led collaborative R&D initiatives.
- Supervised researchers and students, guiding technical development and publications.

March 2019 – Nov 2019 **Part-time Lecturer: Mathematical Statistics**, TSHWANE UNIVERSITY OF TECHNOLOGY, Pretoria, South Africa

- Delivered lectures in Mathematical Statistics and quantitative methods and supported programming and data analysis using Python.
- Supervised and mentored students and assessed quantitative and analytical work.

- Sept 2013 – April 2018 **PhD Student Researcher (Computer Vision and Robotics)**, COUNCIL FOR SCIENTIFIC AND INDUSTRIAL RESEARCH (CSIR), Pretoria, South Africa
- Led doctoral research in computer vision and image processing, addressing open research problems with real-world relevance.
 - Designed and implemented novel algorithms and analytical techniques.
 - Authored technical reports and presented research findings to multidisciplinary stakeholders.
 - Produced and co-authored peer-reviewed publications in journals, books and conference proceedings.
- April 2011 – Aug 2013 **Masters Student Researcher (Image Processing)**, COUNCIL FOR SCIENTIFIC AND INDUSTRIAL RESEARCH (CSIR), Pretoria, South Africa
- Conducted postgraduate research in bio-image processing and analysis.
 - Designed and implemented original image analysis methods and contributed to technical publications.

Selected ML Engineering Projects

- 1 **GenAI Chatbot with RAG for Knowledge Retrieval**, HULAMIN
Developed an enterprise-grade **Generative AI chatbot** integrating a private **LLM** with **Retrieval-Augmented Generation (RAG)**. Built document embedding and semantic-search pipelines using FAISS/ChromaDB over engineering manuals, SOPs and reports. The solution enabled natural-language querying and reduced support requests by 40%.
- 2 **Product Quality Prediction**, HULAMIN
Built a regression-based ML solution linking manufacturing process parameters to product quality. Identified key variables influencing yield consistency and supported proactive operational adjustments.
- 3 **Time Series Forecasting API**, SASOL
Designed and deployed an integrated **Django + React.js** forecasting API combining ARIMA and ML-based models, operationalising model inference through a production application. Enhanced demand planning accuracy and reduced inventory variance by 18%.
- 4 **Predictive Dialing Algorithm**, CONSUMER PROFILE BUREAU
Engineered a predictive ML model optimising agent availability and call distribution in real time, increasing contact-centre efficiency and reducing idle time by 25%.
- 5 **Production ML Deployment Platform**, HULAMIN
Shipped ML applications to production using a **Django/Django REST Framework backend**, **React/TypeScript frontend**, **Docker** containers and **GitHub Actions CI/CD**. Deployed container images through **Azure Container Registry** to **Azure Kubernetes Service (AKS)** and stored trained **.pkl model artifacts in Azure Blob Storage**, creating an end-to-end production path from model artifact to cloud-hosted inference application.

Awards & Recognition

- 2018 **Best Doctor of Engineering Student**, *Council for Scientific and Industrial Research*, Pretoria
- 2018 **Paper Selected for Journal Publication**, *International Joint Conference on Biomedical Engineering Systems and Technologies*, Madeira, Portugal
- 2017 **Best Paper Presentation**, *International Conference on Artificial Intelligence and Evolutionary Computations in Engineering Systems*, Madanapalle, India
- 2014 **Best Master of Engineering Student**, *Council for Scientific and Industrial Research*, Pretoria
- 2009 **Best Bachelor of Science Student**, *University of Limpopo*
- 2019 **Invited Speaker**, *International Conference on Soft Computing and Machine Learning*, Wuhan, China

Communication

Delivered research talks and technical presentations at international AI and engineering conferences.

Languages

English Fluent

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